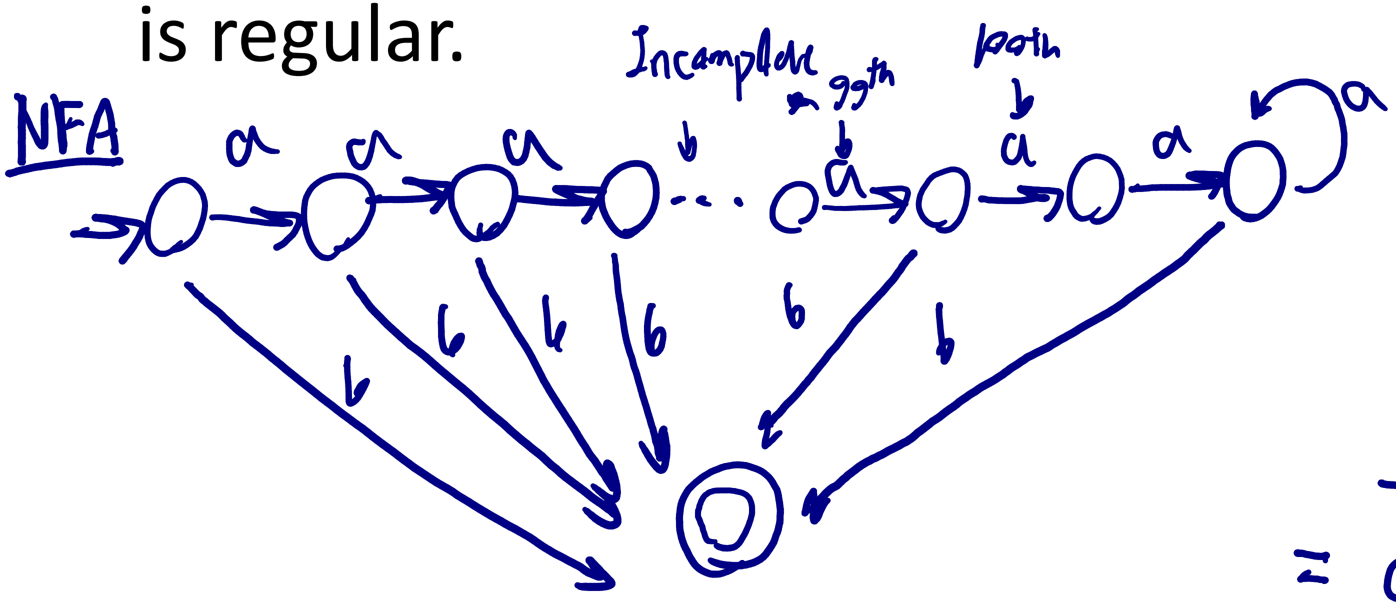
Theory of Computation

Submission link: https://forms.gle/9QfcWiqewGmCcvG36

Exercise 4:
(Closure properties of Regular Language and Regular Expression)

1. Prove that the language $\{a^m b : m \geq 1 \; and \; m \neq 100\}$ is regular.



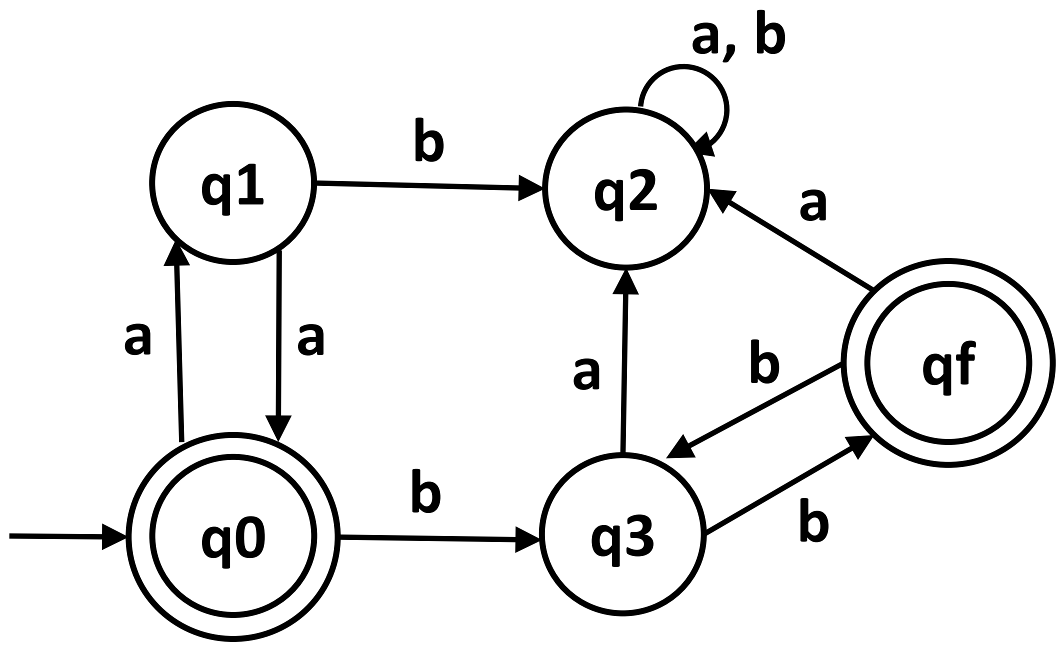
$= \overline{a^{100}b} \cap aa^*b$

2. Find regular expression for the following language

$$L = \{\, w \in \{a, b\}^*: w\ does\ not\ end\ with\ ab \,\}$$

$(a+b)^* \cdot (aa+ba+bb)+\lambda+a+b$

*3. Find regular expression for the following DFA.



$r = (aa)^* \cdot (bb)^*$